CP3 Peer Review: Donut

Reviewed by	Fabricate
Commit reviewed	`0ab1a7c34a4bd2c580b44b444c305310fdd5bb88`

Scores

Item	Score	Reason
Code is maintainable, efficient, and handles errors well	**8/10**	Clean layering, short documented functions, zod validation, and row locks plus transactions around every game change. Main gap: scores aren't checked against the game's state (see the major fault).
Tests are complete for one end-to-end feature and all interfaces of one component, with errors, edge cases and correct assertions	**9/10**	601 back-end tests pass; Games has 109 mocked tests covering every route and error code; the e2e test covers Use Case 4. 23 of 37 interfaces have no no-mock test.
Test code is maintainable and well structured	**8/10**	Clear describe names and shared helpers. `games.test.ts` is 2,577 lines, and the no-mock tests depend on running in order.
Tests match the requirements and design	**8/10**	Tests follow the design's interfaces and Use Case 4. No test enforces Use Case 3's "game has started" precondition, and Testing.md's commit hash and links are stale.
Non-functional requirements are tested well	**3/10**	Targets and designs are documented, but no NFR tests or logs exist yet.
Coverage is high, and anything under 100% is well justified	**7/10**	89% statements, 86% branches overall (47% no-mock only). Coverage screenshots are placeholders and lower-coverage files aren't justified.
All back-end tests can run automatically	**10/10**	GitHub Actions runs every suite against Postgres on each push and PR, and passes.
Codacy runs with the required setup	**10/10**	detekt, ESLint and Trivy enabled with the course configuration, run on every push.
All remaining Codacy issues are well justified	**10/10**	0 issues at the reviewed commit, with no suppressions.

Major fault: one player can set any opponent's rating

Severity: High. It breaks the ELO ranking and leaderboard, two core features, and matchmaking uses the bad ratings to pick opponents.

Details

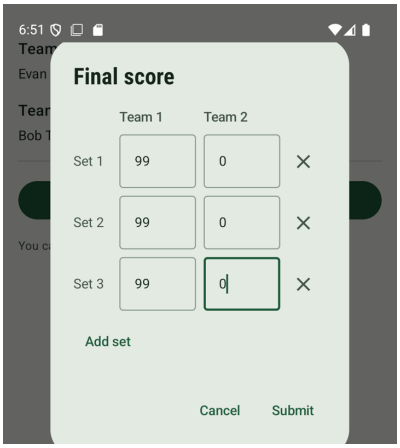
- **No limit on margin of victory.** The update adds 1.5 points per game of margin, summed over all sets, uncapped (`elo.ts:102-105`). The win/loss term is at most 50, so the margin dominates.
- **No tennis score validation.** Any non-negative integer per set and any number of sets are accepted (`games.schemas.ts:12`); the app allows up to 99 per set.
- **One unconfirmed report is final, at any time.** `checkCanEditGame` only rejects completed or cancelled games, so a score is accepted for a matched game nobody accepted, before it starts. Afterwards it can't be accepted, cancelled or re-reported (`409`), so the change is permanent.

How we reproduced it

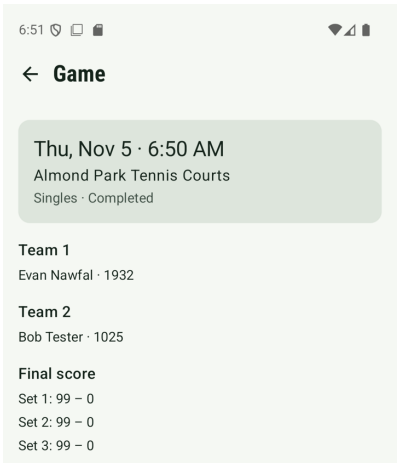
In Donut's own app, built from commit `0ab1a7c` with no code changes (`local.properties` only points it at a local server running the same commit, so no real user's rating was touched). We played as Evan in the app; Bob and Finn are test players acting through the API, exactly as the app would call it.

A. An impossible score is accepted and moves both ratings by 475

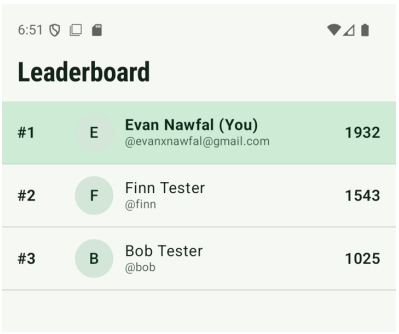
1. Evan (1457) schedules a singles game; Bob (1500) joins team 2, so it is booked.
2. After the start time, Evan taps **Report final score**, enters 99–0, 99–0, 99–0 and submits. The app and server accept it.
3. Evan goes to **1932** (+475) and Bob to **1025** (–475); Evan is now first on the leaderboard.



Step 2: the score form accepts 99–0 sets



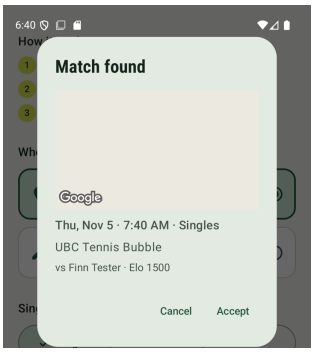
Step 3: game completed, 1932 vs 1025



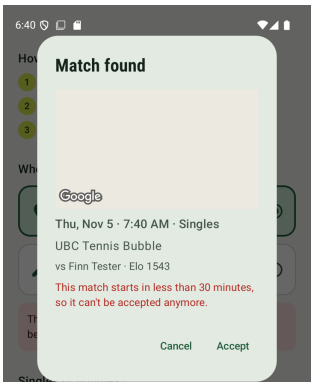
Step 3: leaderboard

B. A player loses rating on a match they never accepted, and can't undo it

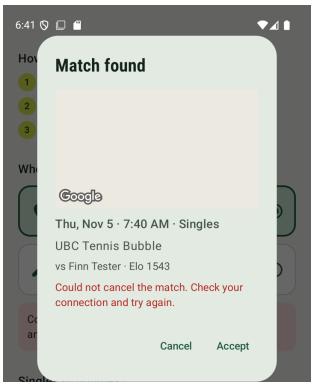
1. Evan and Finn (both 1500) queue for singles. At 6:40, matchmaking proposes a game at 7:40; nobody has accepted.
2. Finn immediately reports 6–0, 6–0 (`POST /games/score`): `200`, the game is completed.
3. In the app, Evan's **Accept** fails with a wrong reason ("starts in less than 30 minutes", at 6:40 for 7:40), and **Cancel** fails.
4. At 6:42, Evan's match history shows a **LOSS, –43** (1500 → 1457) for a 7:40 match that was never accepted or played.



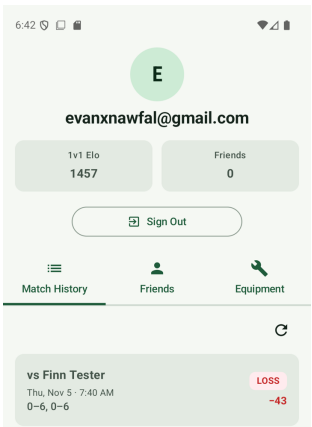
Step 1: 6:40, match at 7:40



Step 3: Accept refused



Step 3: Cancel refused



Step 4: 6:42, loss –43